

Evaluation on Full Dataset for Major Milestones

Baseline

'metrics': {'top_1_hit_rate': 0.07552083333333333, 'top_3_hit_rate': 0.22395833333333334, 'top_5_hit_rate': 0.4296875, 'average_hit_rate': 0.24305555555555558, 'total_scenarios': 384, 'top_1_hits': 29, 'top_3_hits': 86, 'top_5_hits': 165}, 'data_info': {'evaluated_count': 384, 'original_simulation_count': 384, 'original_ground_truth_count': 384}}

Modified Baseline

The evaluation_results is :{'type': 'recommendation', 'metrics': {'top_1_hit_rate': 0.26041666666666667, 'top_3_hit_rate': 0.5078125, 'top_5_hit_rate': 0.6458333333333334, 'average_hit_rate': 0.47135416666666667, 'total_scenarios': 384, 'top_1_hits': 100, 'top_3_hits': 195, 'top_5_hits': 248}, 'data_info': {'evaluated_count': 384, 'original_simulation_count': 384, 'original_ground_truth_count': 384}}

User Summary

The evaluation_results is :{'type': 'recommendation', 'metrics': {'top_1_hit_rate': 0.28125, 'top_3_hit_rate': 0.5390625, 'top_5_hit_rate': 0.6901041666666666, 'average_hit_rate': 0.5034722222222222, 'total_scenarios': 384, 'top_1_hits': 108, 'top_3_hits': 207, 'top_5_hits': 265}, 'data_info': {'evaluated_count': 384, 'original_simulation_count': 384, 'original_ground_truth_count': 384}}

Item Summary

The evaluation_results is :{'type': 'recommendation', 'metrics': {'top_1_hit_rate': 0.265625, 'top_3_hit_rate': 0.5651041666666666, 'top_5_hit_rate': 0.6953125, 'average_hit_rate': 0.5086805555555555, 'total_scenarios': 384, 'top_1_hits': 102, 'top_3_hits': 217, 'top_5_hits': 267}, 'data_info': {'evaluated_count': 384, 'original_simulation_count': 384, 'original_ground_truth_count': 384}}

Engineered Prompt

The evaluation_results is :{'type': 'recommendation', 'metrics': {'top_1_hit_rate': 0.25520833333333333, 'top_3_hit_rate': 0.52864583333333334, 'top_5_hit_rate': 0.765625, 'average_hit_rate': 0.5164930555555556, 'total_scenarios': 384, 'top_1_hits': 98, 'top_3_hits': 203, 'top_5_hits': 294}, 'data_info': {'evaluated_count': 384, 'original_simulation_count': 384, 'original_ground_truth_count': 384}}

User Summary + Item Summary + Engineered Prompt

The evaluation_results is :{'type': 'recommendation', 'metrics': {'top_1_hit_rate': 0.30208333333333333, 'top_3_hit_rate': 0.56770833333333334, 'top_5_hit_rate': 0.7109375, 'average_hit_rate': 0.5269097222222222, 'total_scenarios': 384, 'top_1_hits': 116, 'top_3_hits': 218, 'top_5_hits': 273}, 'data_info': {'evaluated_count': 384, 'original_simulation_count': 384, 'original_ground_truth_count': 384}}

User Summary + Item Summary + Engineered Prompt + Ensemble

The evaluation_results is :{'type': 'recommendation', 'metrics': {'top_1_hit_rate': 0.3098958333333333, 'top_3_hit_rate': 0.6197916666666666, 'top_5_hit_rate': 0.7786458333333334, 'average_hit_rate': 0.5694444444444445, 'total_scenarios': 384, 'top_1_hits': 119, 'top_3_hits': 238, 'top_5_hits': 299}, 'data_info': {'evaluated_count': 384, 'original_simulation_count': 384, 'original_ground_truth_count': 384}}

Evaluation on 100 Task Subset for Ablation Studies

Different Prompt Instructions

No Instruction

The evaluation_results is :{'type': 'recommendation', 'metrics': {'top_1_hit_rate': 0.24, 'top_3_hit_rate': 0.56, 'top_5_hit_rate': 0.7, 'average_hit_rate': 0.5, 'total_scenarios': 100, 'top_1_hits': 24, 'top_3_hits': 56, 'top_5_hits': 70}, 'data_info': {'evaluated_count': 100, 'original_simulation_count': 100, 'original_ground_truth_count': 384}}

User Preference > Candidate Sentiment

The evaluation_results is :{'type': 'recommendation', 'metrics': {'top_1_hit_rate': 0.24, 'top_3_hit_rate': 0.49, 'top_5_hit_rate': 0.66, 'average_hit_rate': 0.46333333333333334, 'total_scenarios': 100, 'top_1_hits': 24, 'top_3_hits': 49, 'top_5_hits': 66}, 'data_info': {'evaluated_count': 100, 'original_simulation_count': 100, 'original_ground_truth_count': 384}}

Candidate Sentiment > User Preference

The evaluation_results is :{'type': 'recommendation', 'metrics': {'top_1_hit_rate': 0.29, 'top_3_hit_rate': 0.68, 'top_5_hit_rate': 0.78, 'average_hit_rate': 0.5833333333333334, 'total_scenarios': 100, 'top_1_hits': 29, 'top_3_hits': 68, 'top_5_hits': 78}, 'data_info': {'evaluated_count': 100, 'original_simulation_count': 100, 'original_ground_truth_count': 384}}

Different Ensemble Sizes

N = 1

The evaluation_results is :{'type': 'recommendation', 'metrics': {'top_1_hit_rate': 0.3, 'top_3_hit_rate': 0.7, 'top_5_hit_rate': 0.82, 'average_hit_rate': 0.6066666666666666, 'total_scenarios': 100, 'top_1_hits': 30, 'top_3_hits': 70, 'top_5_hits': 82}, 'data_info': {'evaluated_count': 100, 'original_simulation_count': 100, 'original_ground_truth_count': 384}}

N = 3

The evaluation_results is :{'type': 'recommendation', 'metrics': {'top_1_hit_rate': 0.37, 'top_3_hit_rate': 0.71, 'top_5_hit_rate': 0.78, 'average_hit_rate': 0.62, 'total_scenarios': 100, 'top_1_hits': 37, 'top_3_hits': 71, 'top_5_hits': 78}, 'data_info': {'evaluated_count': 100, 'original_simulation_count': 100, 'original_ground_truth_count': 384}}

N = 5

The evaluation_results is :{'type': 'recommendation', 'metrics': {'top_1_hit_rate': 0.34, 'top_3_hit_rate': 0.71, 'top_5_hit_rate': 0.76, 'average_hit_rate': 0.6033333333333334, 'total_scenarios': 100, 'top_1_hits': 34, 'top_3_hits': 71, 'top_5_hits': 76}, 'data_info': {'evaluated_count': 100, 'original_simulation_count': 100, 'original_ground_truth_count': 384}}

Different Self-Reflection Voting Strategies and Uncertainties

Self-Reflection

The evaluation_results is :{'type': 'recommendation', 'metrics': {'top_1_hit_rate': 0.25, 'top_3_hit_rate': 0.64, 'top_5_hit_rate': 0.73, 'average_hit_rate': 0.54, 'total_scenarios': 100, 'top_1_hits': 25, 'top_3_hits': 64, 'top_5_hits': 73}, 'data_info': {'evaluated_count': 100, 'original_simulation_count': 100, 'original_ground_truth_count': 384}}

Self-Reflection w/ Greater Uncertainty

The evaluation_results is :{'type': 'recommendation', 'metrics': {'top_1_hit_rate': 0.26, 'top_3_hit_rate': 0.57, 'top_5_hit_rate': 0.7, 'average_hit_rate': 0.5099999999999999, 'total_scenarios': 100, 'top_1_hits': 26, 'top_3_hits': 57, 'top_5_hits': 70}, 'data_info': {'evaluated_count': 100, 'original_simulation_count': 100, 'original_ground_truth_count': 384}}

Self-Reflection w/ Greater Uncertainty Plus 3-way Vote

The evaluation_results is :{'type': 'recommendation', 'metrics': {'top_1_hit_rate': 0.21, 'top_3_hit_rate': 0.57, 'top_5_hit_rate': 0.74, 'average_hit_rate': 0.5066666666666667, 'total_scenarios': 100, 'top_1_hits': 21, 'top_3_hits': 57, 'top_5_hits': 74}, 'data_info': {'evaluated_count': 100, 'original_simulation_count': 100, 'original_ground_truth_count': 384}}

Self-Reflection w/ Greater Uncertainty Plus 2-way Vote

The evaluation_results is :{'type': 'recommendation', 'metrics': {'top_1_hit_rate': 0.24, 'top_3_hit_rate': 0.57, 'top_5_hit_rate': 0.7, 'average_hit_rate': 0.5033333333333333, 'total_scenarios': 100, 'top_1_hits': 24, 'top_3_hits': 57, 'top_5_hits': 70}, 'data_info': {'evaluated_count': 100, 'original_simulation_count': 100, 'original_ground_truth_count': 384}}

Self-Reflection Plus 2-way Vote

The evaluation_results is :{'type': 'recommendation', 'metrics': {'top_1_hit_rate': 0.27, 'top_3_hit_rate': 0.61, 'top_5_hit_rate': 0.74, 'average_hit_rate': 0.54, 'total_scenarios': 100, 'top_1_hits': 27, 'top_3_hits': 61, 'top_5_hits': 74}, 'data_info': {'evaluated_count': 100, 'original_simulation_count': 100, 'original_ground_truth_count': 384}}

Self-Reflection Plus 3-way Vote

The evaluation_results is :{'type': 'recommendation', 'metrics': {'top_1_hit_rate': 0.24, 'top_3_hit_rate': 0.58, 'top_5_hit_rate': 0.71, 'average_hit_rate': 0.5099999999999999, 'total_scenarios': 100, 'top_1_hits': 24, 'top_3_hits': 58, 'top_5_hits': 71}, 'data_info': {'evaluated_count': 100, 'original_simulation_count': 100, 'original_ground_truth_count': 384}}

Different System Modifications for Self-Reflection

Self-Reflection w/ User Summary as Key for Similarity Comparison

The evaluation_results is :{'type': 'recommendation', 'metrics': {'top_1_hit_rate': 0.23, 'top_3_hit_rate': 0.55, 'top_5_hit_rate': 0.74, 'average_hit_rate': 0.5066666666666667, 'total_scenarios': 100, 'top_1_hits': 23, 'top_3_hits': 55, 'top_5_hits': 74}, 'data_info': {'evaluated_count': 100, 'original_simulation_count': 100, 'original_ground_truth_count': 384}}

Self-Reflection w/ User Summary as Key for Similarity Comparison and Fixed LLM Feedback (doesn't mention specifics of past tasks)

The evaluation_results is :{'type': 'recommendation', 'metrics': {'top_1_hit_rate': 0.29, 'top_3_hit_rate': 0.6, 'top_5_hit_rate': 0.74, 'average_hit_rate': 0.5433333333333333, 'total_scenarios': 100, 'top_1_hits': 29, 'top_3_hits': 60, 'top_5_hits': 74}, 'data_info': {'evaluated_count': 100, 'original_simulation_count': 100, 'original_ground_truth_count': 384}}

Self-Reflection w/ User Summary as Key for Similarity Comparison and Fixed LLM Feedback and Structured User Summaries

The evaluation_results is :{'type': 'recommendation', 'metrics': {'top_1_hit_rate': 0.27, 'top_3_hit_rate': 0.62, 'top_5_hit_rate': 0.76, 'average_hit_rate': 0.5499999999999999, 'total_scenarios': 100, 'top_1_hits': 27, 'top_3_hits': 62, 'top_5_hits': 76}, 'data_info': {'evaluated_count': 100, 'original_simulation_count': 100, 'original_ground_truth_count': 384}}